

Nomenclature and Acronyms

For Add-On Course: EE-AOC-581 & EE-AOC-681
Applied Exponential Models for Electrical Engineering - I & II

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About This Document

This document provides a comprehensive list of all technical abbreviations, acronyms, and symbols used throughout the syllabus documents for the add-on course **EE-AOC-581 & EE-AOC-681**: Applied Exponential Models for Electrical Engineering.

How to Use This List

- The list is organized alphabetically.
- Common electrical engineering abbreviations are included alongside specific course-related acronyms.
- For components (like ICs), the full part number and a brief description are provided.

Table 1: Comprehensive List of Nomenclature and Acronyms

Abbreviation	Full Form / Expansion	Description / Context in Course
AICTE	All India Council for Technical Education	The statutory body for technical education in India.
An	Analyse	A cognitive process level in Bloom's Taxonomy.
Ap	Apply	A cognitive process level in Bloom's Taxonomy.
Arduino	Arduino	An open-source electronics platform based on easy-to-use hardware and software. Used for PWM and control implementation.
AY	Academic Year	The period of the academic session (e.g., 2026–27).
BCREC	Dr. B. C. Roy Engineering College	The institution offering the course.
BESS	Battery Energy Storage System	Systems that store energy for later use. Referenced in the context of EV and power systems.
BL	Bloom's Level	Refers to the cognitive process level in Anderson and Krathwohl's Revised Bloom's Taxonomy.
BLDCSim	BLDC Motor Simulation Tool	A specific simulation tool used for BLDC motor analysis in the course.
BLDC	Brushless Direct Current Motor	A synchronous motor powered by a DC source via an inverter. A key application in the course.

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Abbreviation	Full Form / Expansion	Description / Context in Course
Bode Plot	Bode Plot	A frequency response plot (magnitude and phase) used to analyze system stability.
B.Tech	Bachelor of Technology	The undergraduate degree program.
C	Capacitor	A passive electrical component. The current-second balance property is a key concept.
CCM	Continuous Conduction Mode	A mode of operation for SMPS where the inductor current never falls to zero.
CO	Course Outcome	Measurable statements describing what a student will know or be able to do after completing the course.
Cr	Create	The highest cognitive process level in Bloom's Taxonomy.
CRO	Cathode Ray Oscilloscope	A tool for measuring and displaying electrical waveforms.
CTR	Current Transfer Ratio	A parameter of an optocoupler, the ratio of output current to input current.
D	Duty Cycle	The fraction of one period during which a signal is active (high).
DCM	Discontinuous Conduction Mode	A mode of operation for SMPS where the inductor current falls to zero during a part of the switching cycle.
dq0 / dq	Direct-Quadrature-Zero	A mathematical transformation (Park's transform) used to simplify AC machine analysis.
d-axis	Direct Axis	The flux-producing axis in the dq0 reference frame.
EE	Electrical Engineering	The core department for this course.
ESR	Equivalent Series Resistance	The internal resistance of a capacitor, which introduces a zero in the transfer function.
ESP32	ESP32	A low-cost, low-power system on a chip (SoC) with Wi-Fi and Bluetooth capabilities. Used for advanced PWM and IoT.
EV	Electric Vehicle	A vehicle powered by electric motors. A major application area for the course content.
Ev	Evaluate	A cognitive process level in Bloom's Taxonomy.
FOC	Field-Oriented Control	A motor control strategy that controls the stator currents in a rotating reference frame (dq), analogous to a DC motor.
GaN	Gallium Nitride	A wide-bandgap semiconductor used in high-performance power electronics.
$G_{vd}(s)$	Control-to-Output Transfer Function	The transfer function of a converter from the control input (duty cycle) to the output voltage.
HIL	Hardware-in-the-Loop	A simulation technique where a physical controller is connected to a simulated plant.
Hrs / Hours	Hours	Contact hours for each module.
I_C	Capacitor Current	The current flowing through a capacitor.
i_d	d-axis Current	The flux-producing current component in a motor.
i_q	q-axis Current	The torque-producing current component in a motor.

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Abbreviation	Full Form / Expansion	Description / Context in Course
IDE	Integrated Development Environment	A software application that provides comprehensive facilities to programmers.
IoT	Internet of Things	The network of physical objects embedded with sensors and software.
KCL	Kirchhoff's Current Law	A fundamental law for current in electrical circuits.
KVL	Kirchhoff's Voltage Law	A fundamental law for voltage in electrical circuits.
K-Factor	K-Factor Method	A systematic method for designing Type-II and Type-III compensators for SMPS.
L	Inductor	A passive electrical component. The volt-second balance property is a key concept.
LED	Light-Emitting Diode	Used for indicators and as part of the gate drive circuits.
LHP	Left-Half Plane	The left side of the complex plane. Poles in the LHP indicate stability.
LM317	LM317	A common 3-terminal positive adjustable linear voltage regulator.
LM337	LM337	A common 3-terminal negative adjustable linear voltage regulator.
LTspice	LTspice	A high-performance SPICE simulation software, used for power electronics circuit simulation.
NAAC	National Assessment and Accreditation Council	A body that accredits higher education institutions in India.
NBA	National Board of Accreditation	A body that accredits technical programs (like B.Tech) in India.
NEP 2020	National Education Policy 2020	India's new education policy framework.
NMP	Non-Minimum Phase	A system with zeros in the right-half plane. The Boost converter is an example.
NumPy	Numerical Python	A Python library for numerical computing.
OBE	Outcome-Based Education	An educational approach focused on defining and measuring student outcomes.
OpenModelica	OpenModelica	An open-source Modelica-based modeling and simulation environment.
ORCID	Open Researcher and Contributor ID	A non-proprietary alphanumeric code to uniquely identify scientific authors.
Oscilloscope	Oscilloscope	An electronic test instrument that displays varying signal voltages.
Park Transform	Park Transformation	The abc-to-dq0 transformation.
PC817	PC817	A common 4-pin DIP optocoupler IC.
PID	Proportional-Integral-Derivative Controller	A generic feedback control loop mechanism. The course focuses on Type-II/III instead, as they are purpose-built for SMPS.

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Abbreviation	Full Form / Expansion	Description / Context in Course
PIL	Processor-in-the-Loop	A simulation technique where the controller code is run on the target processor, while the plant is simulated.
PMSM	Permanent Magnet Synchronous Motor	A popular motor type for EV and industrial drives. A main application for FOC.
PO	Programme Outcome	Statements that describe what students are expected to know or be able to do by the time of graduation.
PSO	Programme Specific Outcome	Statements specific to a particular program (e.g., B.Tech EE).
PWM	Pulse Width Modulation	A technique to encode a signal's amplitude into a pulse train's duty cycle. Core topic for gate drive.
Python	Python	A high-level programming language used for analysis and plotting.
q-axis	Quadrature Axis	The torque-producing axis in the dq0 reference frame.
R	Resistor	A passive electrical component. Its behavior is fundamental to circuit analysis.
RC	Resistor-Capacitor	A type of circuit used for filtering and timing.
Re	Remember	A cognitive process level in Bloom's Taxonomy.
RHP	Right-Half Plane	The right side of the complex plane. Zeros in the RHP cause non-minimum phase behavior.
RL	Resistor-Inductor	A type of circuit used for filtering and energy storage.
R_G	Gate Resistance	A resistor placed in series with the gate of a power transistor to control switching speed and EMI.
Scilab	Scilab	A free and open-source numerical computation software.
SiC	Silicon Carbide	A wide-bandgap semiconductor used in high-performance power electronics.
SMPS	Switch Mode Power Supply	A power supply that uses a switching regulator to convert electrical power efficiently.
SoC	State of Charge	The level of charge of an electric battery, expressed as a percentage.
SPWM	Sinusoidal Pulse Width Modulation	A modulation technique used in motor drives to generate sinusoidal voltage waveforms.
SVPWM	Space Vector Pulse Width Modulation	An advanced modulation technique for motor drives, offering better DC bus utilization.
TI	Texas Instruments	A major semiconductor manufacturer.
Type-II	Type-II Compensator	A compensator with one zero and two poles, used for stabilizing SMPS.
Type-III	Type-III Compensator	A compensator with two zeros and three poles, used for stabilizing SMPS with more complex plants.
T	Torque	The rotational force produced by an electric motor.
UCC27524	UCC27524	A TI dual 5A gate driver IC for power MOSFETs and IGBTs.
Un	Understand	A cognitive process level in Bloom's Taxonomy.
UVLO	Under-Voltage Lock-out	A safety feature of power ICs that disables the output when the supply voltage is too low.
V_{in}	Input Voltage	The input voltage to a converter.

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Abbreviation	Full Form / Expansion	Description / Context in Course
V_o	Output Voltage	The output voltage from a converter.
V_L	Inductor Voltage	The voltage across an inductor.
WEBENCH	WEBENCH Power Designer	A TI online tool for designing power supplies.
Xcos	Xcos	A Scilab package for modeling and simulation of dynamical systems.
7805	7805	A 5V fixed positive linear voltage regulator.
7812	7812	A 12V fixed positive linear voltage regulator.
7905	7905	A -5V fixed negative linear voltage regulator.
6N137	6N137	A high-speed 10 Mbit/s optocoupler IC.
555 Timer	555 Timer	An integrated circuit used for timer, pulse generation, and oscillator applications.

Mathematical and Symbol Notation

- e : Euler's number (approx. 2.71828).
- e^t : Pure exponential growth.
- e^{-t} : Pure exponential decay.
- $1 - e^{-t}$: Saturating rise.
- $\tanh$: Hyperbolic tangent.
- $\cosh$: Hyperbolic cosine (catenary).
- $e^{j\omega t}$: A complex exponential representing a pure sinusoidal (phasor).
- $e^{(\sigma+j\omega)t}$: A damped complex exponential, where σ is the damping factor and ω is the angular frequency.
- $s = \sigma + j\omega$: Complex frequency variable.
- $\int V_L dt = 0$: Volt-second balance.
- $\int I_C dt = 0$: Current-second balance.
- $\frac{V_o}{V_{in}} = D$: Buck converter gain.
- $\frac{V_o}{V_{in}} = \frac{1}{1-D}$: Boost converter gain.